

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial 0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

Passive interfaces impact neighbour relationship to break down making the other routers can't learn the routes through this routers.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2 (config)# router ospf 1

R2 (config-router)# no passive-interface S0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial 0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 → serial link cost 64 + ethernet cost 1

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because 100 Mb/s default bandwidth is not sufficient to differentiate between Gb links. Changing the reference bandwidth allow OSPF cost to reflect the actual speed of each link more accurately.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

$192.168.3.0/24 \rightarrow 781 + 1 = 782$ [R1 S0/0/1 + R3 G0/0]

$192.168.23.0/30 \rightarrow 781 + 64 = 845$ [R1 S0/0/1 + R3 S0/0/1]

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562 because the route to 192.168.23.0/24 is using 2 serial links (781 each) $\rightarrow 782 \times 2 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The `ip ospf cost 1565` command sets the cost of R1 S0/0/1 to 1565. This makes it more costly to go directly through this interface than the path through R2. It causes OSPF to choose the lower cost path through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because in OSPF, router ID should be a unique identifier. If the ID is clashing, it may lead to a confusing routing information.

2. Why is the DR/BDR election process not a concern in this lab?

Because DR/BDR will be related to this lab only in the case of multi-access networks (e.g., Ethernet or Frame Relay). Therefore, as the serial links used in this lab is point-to-point, we don't have to worry about DR/BDR election here.

3. Why would you want to set an OSPF interface to passive?

Because we want the interface to stop sending OSPF Hello packets and stop establishing adjacencies, which reduces unnecessary routing update traffic and avoid leaking routing information on LANs which is not required the dynamic routing updates.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of Areas in OSPF: separating the network into smaller areas, and the layered design help improve the scalability of the network.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

```
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
```

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

Since the ~~OSPF~~ route has a lower administrative distance of 110 (RIP is 120, and EIGRP is 90 internal and 170 external), it will be selected as a preferred route.

EIGRP

0